

Unit 2, Lesson 1: Square Values



Benchmark

MA.8.AR.2.3 Given an equation in the form of $x^2 = p$ and $x^3 = q$ where p is a whole number and q is an integer, determine the real solutions.



Learning Target

- ☐ I can use multiple representations to explore square values.



2.1.1 Warm-Up: Reflecting on Learning

Learning something new can sometimes be challenging, but the challenges are part of what leads to success.

"Failure is not the opposite of success. It is part of success." -Arianna Huffington

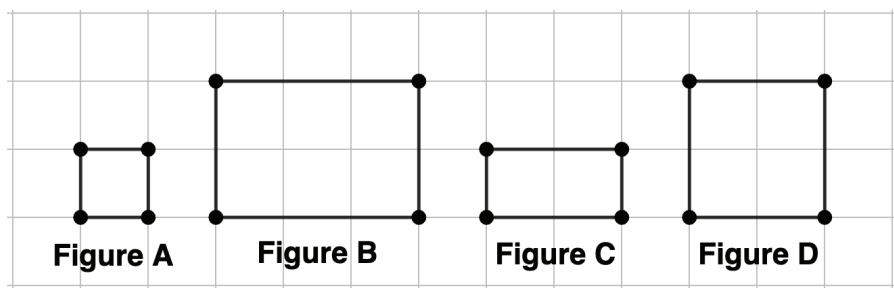
1. What would you want your teacher to say to you when they see you taking a risk, trying hard, or pushing through mistakes to learn something new?
2. Describe how it feels when you learn or accomplish something that is really challenging.



2.1.2 Exploration: Area of Squares

Complete the following with your partner.

- Four figures are shown on the grid, where each gridline is 1 unit.



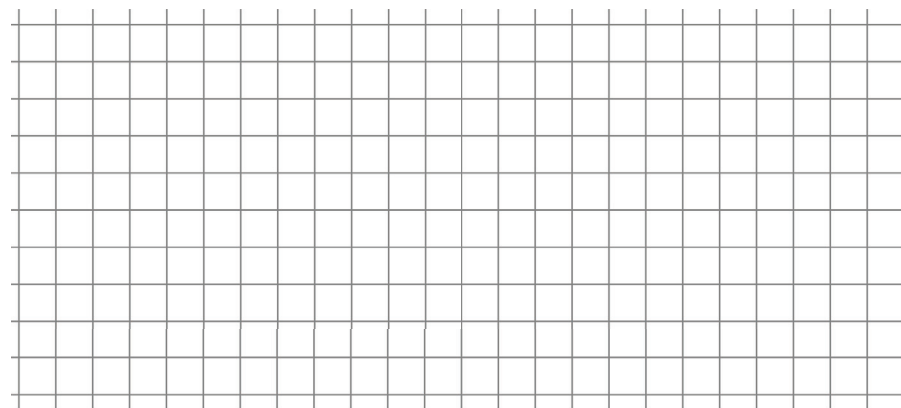
- Find the dimensions and area of each figure.

Figure	Dimensions	Area (units ²)
A		
B		
C		
D		

- All of the figures above are rectangles. Explain, using the dimensions, which of the shapes are also squares.

- With your partner, create squares whose areas range from 1 to 25 square units. Create as many as possible on the grid paper below. Each gridline has a length of 1 unit.

- For each square, label the length and the width, along with the area inside.



- Form a group with another pair of partners. Compare the squares each group created and, if necessary, add any squares to your grid.

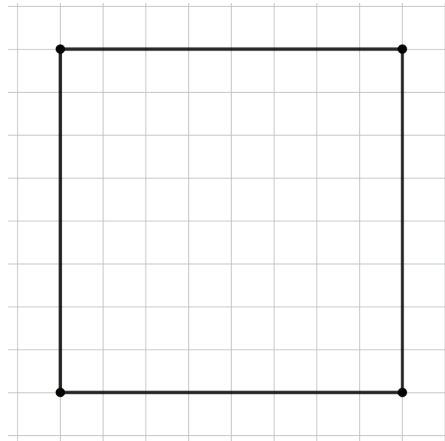
- Alicia says, "1 squared is 1 and 2 squared is 4." Explain what you think it means to square a number.



2.1.3 Bring It Together: Area of Squares

1. The area of the square shown is 64 square units.

- a. Label the length and the width of the square on the grid.



- b. The length and width can be multiplied to find the area of 64 square units. Since the figure is a square, the area can also be found using $A = s^2$, where s is the length of a side.

Complete the equations for the square shown.

$$\text{Area} = (\underline{\hspace{1cm}})^2 \quad \text{or} \quad \text{Area} = \underline{\hspace{1cm}} \times \underline{\hspace{1cm}}$$

2. Consider a square with an area of 100 units².

- a. What is the length of each side?
- b. Explain how you determined the length.

3. Consider a rectangle with an area of 8 square units.

- a. Use the grid to draw all of the possible rectangles with an area of 8 square units.



- b. Explain whether it is possible to have a square with an area of 8 square units.

4. The numbers 1, 4, 16, 25, 64, and 100 are all examples of perfect squares. The numbers 3, 8, and 200 are nonexamples of perfect squares.

- a. Discuss with your partner what it means for a number to be a perfect square. Then, write your definition of a perfect square.
- b. Ask two other pairs of partners the definition they wrote. Listen to their definition and write a summary. *You are the only person who should write in the first column of the table.* Have the person initial next to your summary stating that your summary was accurate.

My Summary of the Definition of a Perfect Square	Initials

- c. Review the definitions with your partner and make any necessary adjustments to your own definition.



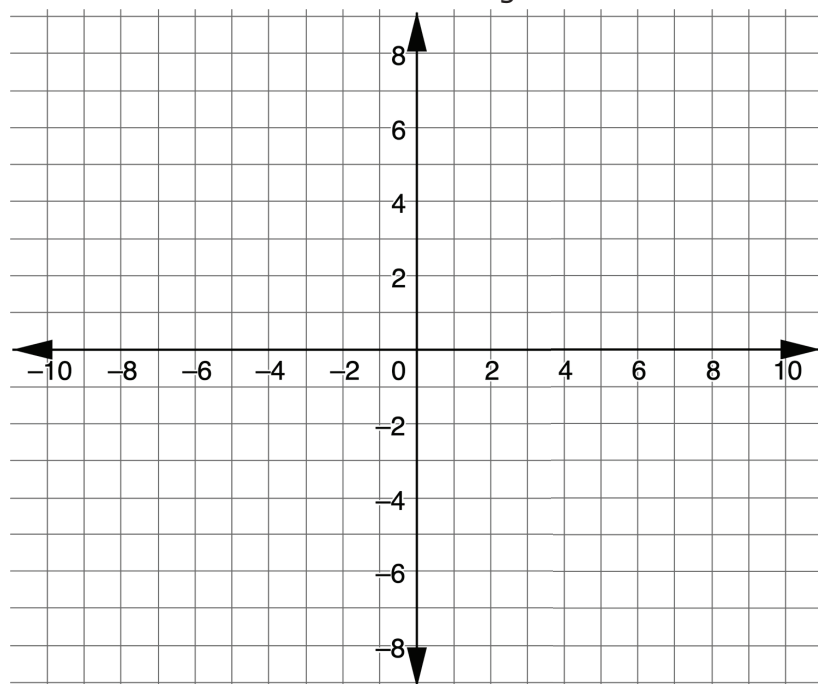
2.1.4 Exploration: Square Values on the Coordinate Plane

1. Use the equation $y = x^2$ to complete the following with your partner.
- a. Complete the table of values for the equation $y = x^2$ by substituting the given values of x and solving for the corresponding y -value.

x	Substitute into $y = x^2$	Coordinates (x, y)
-3	$y = (-3)^2$ $y = \underline{\hspace{2cm}}$	(-3, $\underline{\hspace{2cm}}$)
-2		
-1		
0		
1		
2		
3		

- b. Discuss with your partner why the y -values are all perfect squares. Summarize your discussion.

c. Plot the coordinates on the grid shown.



d. The graph of $y = x^2$ does not go below the x -axis. Complete the statements.

A positive number multiplied by itself results in a

- ☐ positive
- ☐ negative

product. A negative number

multiplied by itself results in a

- ☐ positive
- ☐ negative

product. Therefore, for any value of x multiplied by

itself other than 0, the value of y will

- ☐ sometimes
- ☐ always

be

- ☐ positive,
- ☐ negative,

so all of the points on the graph

of $y = x^2$ other than $(0,0)$ are above the x -axis.

2. Four coordinate pairs are shown. The points represented are also on the graph of $y = x^2$.

$$(1.5, 2.25) \quad (-1.5, 2.25) \quad \left(\frac{5}{2}, \frac{25}{4}\right) \quad \left(-\frac{5}{2}, \frac{25}{4}\right)$$

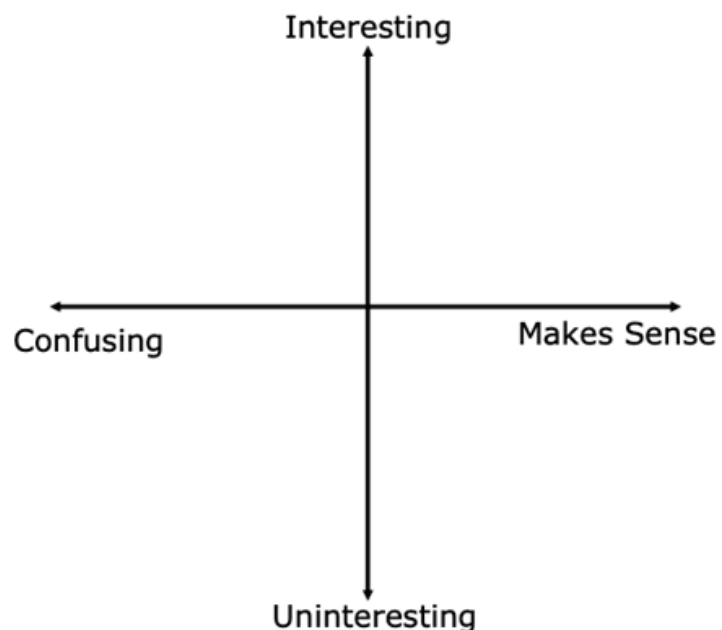
- a. Locate and plot the given points on the graph.
- b. Discuss with your partner what you notice about the given coordinate pairs. Then, summarize your discussion.
- c. Choose one of the points and show how it satisfies the equation $y = x^2$.





2.1.5 Wrap-Up: Reflection

1. Plot a point on the grid below to indicate your level of interest and understanding of today's learning targets.



2. Write one question you have about square values after today's lesson.

Unit 2, Lesson 2: Square Values and Square Roots



Benchmarks

MA.8.NSO.1.1 Extend previous understanding of rational numbers to define irrational numbers within the real number system. Locate an approximate value of a numerical expression involving irrational numbers on a number line.

MA.8.AR.2.3 Given an equation in the form of $x^2 = p$ and $x^3 = q$, where p is a whole number and q is an integer, determine the real solutions.



Learning Targets

- ☐ I can determine the positive and negative solutions for $x^2 = p$, where p is a whole number.
- ☐ I can use the vocabulary related to radicals.
- ☐ I can interpret and evaluate a square root.